

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0789

Castle Hills 100 kV
May 16, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0789. The Interconnection Customer proposes a 151.9 MW solar photovoltaic facility interconnecting at the Castle Hills 100 kV substation in the Indiana area. The request was received on April 29, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 237 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,541,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: S HAVEN - ANDERSON 100.0 kV L
Contingency: ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV Transformer Bank
Loading: 231.1% (Rating: 113 MVA)
DFAX: 100.0%
Pre-project loading: 96.4%
2. Facility: SVONA - RICHARD LAKE 100.0 kV line
Contingency: E METHUEN 500/230 kV Autotransformer
Loading: 172.9% (Rating: 110 MVA)
DFAX: 5.3%
Pre-project loading: 165.5%
3. Facility: BUTLER MUNICIPAL POWER PLANT - SAVONA 100.0 kV
Contingency: EASTMETHUEN 500/230kV Transformer
Loading: 169.9% (Rating: 121 MVA)
DFAX: 5.3%
Pre-project loading: 163.2%
4. Facility: CASTLE HILLS - ANDERSON 100.0 kV
Contingency: WALLENPAUPACK - MINNEAPOLIS CITY OF 500.0 kV
Loading: 163.6% (Rating: 138 MVA)
DFAX: 100.0%
Pre-project loading: 53.6%
5. Facility: RICHARD LAKE - CARBONDALE NW 100.0 kV
Contingency: E MEHUEN 500-230 kV Auto
Loading: 144.3% (Rating: 116 MVA)
DFAX: 5.3%
Pre-project loading: 137.3%

6. Facility: HANSEL - BUTLER MUNICIPAL POWER PLANT 100.0 kV
Contingency: E METHUEN 500/230 kV Transformer Bank
Loading: 142.1% (Rating: 155 MVA)
DFAX: 5.3%
Pre-project loading: 136.9%
7. Facility: HANSEL - THOUSAND SPRINGS SUBSTATION 100.0 kV
Contingency: E METHUEN 500/230 kV Auto
Loading: 138.6% (Rating: 165 MVA)
DFAX: 5.3%
Pre-project loading: 133.7%
8. Facility: LOOKINGGLASS - TRUMAN 230.0 kV L
Contingency: TREGO - BEAVER 500.0 kV
Loading: 126.3% (Rating: 575 MVA)
DFAX: 6.5%
Pre-project loading: 124.5%
9. Facility: SEMASS TAP - ANCHOR HOCKING 100.0 kV
Contingency: POWERLINE ELKHORN - WEBSTERVILLE 100.0 kV line
Loading: 126.0% (Rating: 116 MVA)
DFAX: 5.3%
Pre-project loading: 119.1%
10. Facility: SNYDER - SEMASS TAP 100.0 kV
Contingency: WEBSTERVILLE - POWERLINE ELKHORN 100.0 kV
Loading: 123.8% (Rating: 113 MVA)
DFAX: 5.3%
Pre-project loading: 116.7%
11. Facility: RMEC - S HAVEN 100.0 kV
Contingency: CANAL ST - LOWER GRANITE 100.0 kV line
Loading: 116.5% (Rating: 120 MVA)
DFAX: 27.6%
Pre-project loading: 81.6%
12. Facility: GALION GENERATING STA 500 kV/230 kV XFMR
Contingency: TREGO - BEAVER 500.0 kV line
Loading: 116.4% (Rating: 1,059 MVA)
DFAX: 8.1%
Pre-project loading: 115.2%
13. Facility: GEORGE NEAL S - MYRTLE POINT 230.0 kV
Contingency: SWAN VLY - TREGO 500.0 kV
Loading: 115.9% (Rating: 670 MVA)
DFAX: 8.0%
Pre-project loading: 114.1%
14. Facility: EUFAULA - BEAVER 500.0 kV Line
Contingency: TREGO SUBSTATION - GALION GENERATING STA 500.0 kV Line
Loading: 115.9% (Rating: 3,064 MVA)
DFAX: 23.7%
Pre-project loading: 114.7%
15. Facility: ZONA LIBRE 6007 - GEORGE NEAL SOUTH 230.0 kV
Contingency: SWAN VLY - TREGO 500.0 kV
Loading: 115.5% (Rating: 694 MVA)
DFAX: 8.0%
Pre-project loading: 113.8%
16. Facility: RICHLAND - TROY ENERGY LLC 500.0 kV
Contingency: TREGO - SWAN VALLEY 500.0 kV L

- Loading: 115.5% (Rating: 2,642 MVA)
DFAX: 11.7%
Pre-project loading: 114.8%
17. Facility: ZONA LIBRE 6007 - KAMPSVILLE 230.0 kV
Contingency: SWAN VLY - TREGO 500.0 kV
Loading: 115.1% (Rating: 719 MVA)
DFAX: 8.0%
Pre-project loading: 113.5%
18. Facility: MORENO TAP - KAMPVILLE 230.0 kV
Contingency: SWAN VALLEY - TREGO 500.0 kV
Loading: 114.9% (Rating: 733 MVA)
DFAX: 8.0%
Pre-project loading: 113.3%
19. Facility: EASTOWN RD - MORENO TAP 230.0 kV
Contingency: SWAN VLY - TREGO 500.0 kV
Loading: 114.6% (Rating: 758 MVA)
DFAX: 8.0%
Pre-project loading: 113.0%
20. Facility: BURRIS PIT - HEBO 230.0 kV
Contingency: TREGO - BEAVER 500.0 kV
Loading: 114.3% (Rating: 777 MVA)
DFAX: 10.6%
Pre-project loading: 112.2%
21. Facility: CORRIDOR - LOOKINGGLASS 230.0 kV line
Contingency: TREGO - BEAVER 500.0 kV
Loading: 114.3% (Rating: 608 MVA)
DFAX: 6.5%
Pre-project loading: 112.6%
22. Facility: BURRIS PIT - JAMES DE YOUNG 230.0 kV
Contingency: TREGO - BEAVER 500.0 kV
Loading: 113.5% (Rating: 801 MVA)
DFAX: 10.6%
Pre-project loading: 111.5%
23. Facility: GRAY'S FLAT - E METHUEN 230.0 kV L
Contingency: MADELINE - E METHUEN SUBSTATION 230.0 kV
Loading: 113.2% (Rating: 808 MVA)
DFAX: 12.6%
Pre-project loading: 110.8%
24. Facility: GALION GENERATING STATION - MYRTLE POINT 230.0 kV line
Contingency: TREGO - BEAVER 500.0 kV
Loading: 111.7% (Rating: 1,011 MVA)
DFAX: 5.8%
Pre-project loading: 110.8%
25. Facility: ELDEAN - CARBONDALE NW 100.0 kV Line
Contingency: EAST METHUEN 500/230 kV Transformer Bank
Loading: 111.4% (Rating: 129 MVA)
DFAX: 5.3%
Pre-project loading: 105.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S HAVEN - ANDERSON 100.0 kV L	Rebuild line with higher-capacity conductor	\$7,074,000	57% (\$4,047,000)
SVONA - RICHARD LAKE 100.0 kV line	Rebuild line with higher-capacity conductor	\$9,459,000	12% (\$1,178,000)
BUTLER MUNICIPAL POWER PLANT - SAVONA 100.0 kV	Rebuild line with higher-capacity conductor	\$1,855,000	12% (\$231,000)
CASTLE HILLS - ANDERSON 100.0 kV	Rebuild line with higher-capacity conductor	\$5,182,000	100% (\$5,182,000)
RICHARD LAKE - CARBONDALE NW 100.0 kV	Rebuild line with higher-capacity conductor	\$3,529,000	12% (\$440,000)
HANSEL - BUTLER MUNICIPAL POWER PLANT 100.0 kV	Rebuild line with higher-capacity conductor	\$5,671,000	12% (\$707,000)
HANSEL - THOUSAND SPRINGS SUBSTATION 100.0 kV	Rebuild line with higher-capacity conductor	\$5,499,000	12% (\$685,000)
LOOKINGGLASS - TRUMAN 230.0 kV L	Rebuild line with higher-capacity conductor	\$22,501,000	2% (\$379,000)
SEMASS TAP - ANCHOR HOCKING 100.0 kV	Rebuild line with higher-capacity conductor	\$7,917,000	10% (\$811,000)
SNYDER - SEMASS TAP SUBSTATION 100.0 kV L	Rebuild line with higher-capacity conductor	\$3,808,000	10% (\$390,000)
RMEC - S HAVEN 100.0 kV	Rebuild line with higher-capacity conductor	\$3,423,000	26% (\$902,000)
GALION GENERATING STA 500-230 kV Auto	Replace transformer with higher-rated unit	\$22,543,000	2% (\$415,000)
MYRTLE POINT SUBSTATION - GEORGE NEAL S 230.0 kV line	Rebuild line with higher-capacity conductor	\$18,469,000	3% (\$533,000)
EUFAULA - BEAVER 500.0 kV Line	Rebuild line with higher-capacity conductor	\$48,483,000	2% (\$901,000)
ZONA LIBRE 6007 - GEORGE NEAL SOUTH 230.0 kV	Rebuild line with higher-capacity conductor	\$41,022,000	3% (\$1,184,000)
RICHLAND - TROY ENERGY LLC 500.0 kV	Rebuild line with higher-capacity conductor	\$181,751,000	1% (\$2,210,000)
ZONA LIBRE 6007 - KAMPSVILLE 230.0 kV	Rebuild line with higher-capacity conductor	\$43,413,000	3% (\$1,253,000)
MORENO TAP - KAMPSVILLE 230.0 kV	Rebuild line with higher-capacity conductor	\$49,314,000	3% (\$1,423,000)
EASTOWN RD - MORENO TAP 230.0 kV	Rebuild line with higher-capacity conductor	\$54,523,000	3% (\$1,574,000)
HEBO SUBSTATION - BURRIS PIT 230.0 kV	Rebuild line with higher-capacity conductor	\$24,726,000	2% (\$597,000)
CORRIDOR - LOOKINGGLASS 230.0 kV line	Rebuild line with higher-capacity conductor	\$29,201,000	2% (\$492,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BURRIS PIT - JAMES DE YOUNG 230.0 kV	Rebuild line with higher-capacity conductor	\$60,096,000	2% (\$1,450,000)
GRAY'S FLAT - EAST METHUEN SUBSTATION 230.0 kV	Rebuild line with higher-capacity conductor	\$42,474,000	2% (\$886,000)
GALION GENERATING STATION - MYRTLE POINT 230.0 kV line	Rebuild line with higher-capacity conductor	\$18,031,000	2% (\$340,000)
ELDEAN - CARBONDALE NW 100.0 kV Line	Rebuild line with higher-capacity conductor	\$2,656,000	12% (\$331,000)
	Total allocated to project		\$28,541,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.